

Part C: Data Request

Purpose of the request

To commence the study, we require historical data that will allow us to understand and predict late or unsuccessful delivery attempts. The request is intentionally detailed so that data availability, quality and potential gaps can be identified early. Where possible, please provide the raw records rather than aggregated reporting outputs.

Preferred historical period: 2–4 years of delivery and logistics history, including successful and unsuccessful delivery attempts. Please provide anonymised customer, driver and vehicle identifiers where required

1. Orders & Deliveries

Column name	Data type	Example	Additional information
Order ID	String	ORD123456	Unique identifier for each order.
Order date	Datetime	2025-05-03 09:15:00	Date and time the order was placed.
Dispatch date	Datetime	2025-05-04 07:30:00	Date and time the order left the fulfilment location.
Promised delivery	Datetime	2025-05-05 18:00:00	Date and time promised to the customer.
Delivery date	Datetime	2025-05-05 17:42:00	Actual successful delivery date and time.
Delivery status	Categorical	Delivered	Include all possible statuses, e.g. Delivered, Late, Failed, Cancelled, Returned.
Delivery attempt	Integer	1	Number of the delivery attempt.
Delivery outcome	Categorical	Successful	Outcome of each attempt, e.g. Successful, Customer unavailable, Incorrect address, Refused, Other.
Delivery duration	Numeric	42	Duration in minutes from start to completion of the delivery attempt.
Late delivery	Binary	0	Target variable. 1 = delivered after promised time; 0 = on time.
Repeat delivery	Binary	0	1 = required more than one delivery attempt; 0 = first attempt successful.
Order value	Numeric	1299.99	Total order value.
Item count	Integer	3	Number of items in the order.
Product category	Categorical	Electronics	Main product category. Include a consistent category list.
Delivery type	Categorical	Standard	E.g. Standard, Express, Same-day.
Customer ID	String	CUST45678	Anonymised unique customer identifier.

2. Delivery & Route Data

Column name	Data type	Example	Additional information
Delivery ID	String	DEL98765	Unique identifier for each delivery attempt.
Route ID	String	RTE1023	Identifier for the route used.
Driver ID	String	DRV456	Anonymised driver identifier.
Vehicle ID	String	VEH789	Anonymised vehicle identifier.
DC ID	String	DC03	Distribution centre from which the order was dispatched.
Delivery area	String, categorical	Durban North	Delivery area/suburb.
Delivery latitude	Numeric	-29.755	Customer delivery latitude, where available.
Delivery longitude	Numeric	31.041	Customer delivery longitude, where available.
Route distance	Numeric	18.6	Planned distance in kilometres.
Actual distance	Numeric	21.4	Actual distance travelled in kilometres.
Planned duration	Numeric	45	Expected route/delivery duration in minutes.
Actual duration	Numeric	58	Actual duration in minutes.
Stop number	Integer	7	Position of the delivery within the route.
Route start	Datetime	2025-05-05 07:00:00	Time the route started.
Delivery window	String/categorical	14:00–18:00	Customer's promised/selected delivery window.
Traffic condition	Categorical	Heavy	If available; otherwise provide underlying traffic data.
Weather condition	Categorical	Rain	If available for the delivery location/time.

3. Customer & Delivery History

Column name	Data type	Example	Additional information
Customer tenure	Numeric	18	Number of months since the customer's first order.
Previous orders	Integer	24	Number of previous orders before the current order.
Previous failures	Integer	2	Number of previous failed delivery attempts.
Previous late deliveries	Integer	3	Number of previous late deliveries.
Preferred delivery time	Categorical	Afternoon	If available.
Address type	Categorical	Residential	E.g. Residential, Business, Other.
Previous address issues	Integer	1	Number of previous deliveries affected by address problems.

4. Driver & Vehicle Data

Column name	Data type	Example	Additional information
Driver experience	Numeric	14	Months employed/driving for STADIOalot.
Driver deliveries	Integer	1830	Number of deliveries completed historically.
Driver late rate	Numeric	0.08	Historical late-delivery rate calculated using information available before the current delivery.
Driver failed rate	Numeric	0.04	Historical failed-delivery rate.
Vehicle type	Categorical	Van	E.g. Van, Truck, Motorcycle.
Vehicle capacity	Numeric	1200	Capacity in kg or another consistent unit.
Vehicle load	Numeric	850	Load assigned to the vehicle for the route.

5. Operational & Time Data

Column name	Data type	Example	Additional information
Day of week	Categorical	Monday	Derived from order/delivery timestamp if not already available.
Month	Integer	5	1–12.
Public holiday	Binary	0	1 = public holiday; 0 = not a public holiday.
Peak period	Binary	1	Identify major sales/peak periods such as Black Friday if available.
Orders on route	Integer	24	Total deliveries assigned to the route.
Orders per driver	Integer	24	Number of orders assigned to the driver for the shift.
Warehouse processing time	Numeric	38	Time in minutes from order readiness to dispatch, if available.

6. Data Quality & Reference Information

Column name	Data type	Example	Additional information
Record timestamp	Datetime	2025-05-05 18:05:00	When the delivery event was recorded.
Data source	Categorical	Delivery system	System/table from which the record originated.
Missing reason	String/categorical	Not captured	Explain why important fields are missing where possible.
Cancellation reason	Categorical	Customer cancelled	Required for cancelled orders.
Return reason	Categorical	Customer unavailable	Include standardised return/reason codes.

Data Format Requirements

Please provide the **raw historical data rather than only aggregated KPIs**. Ideally, we would receive at least **2–4 years of historical delivery records**, including both successful and unsuccessful deliveries. The briefing confirms that STADIOalot has four years of delivery/logistics history available.

Expected Outcomes

Our final and important data-science requirement is that variables such as **driver historical failure rate, customer history and route performance must be calculated using information available before the delivery being predicted**. Otherwise, information from the outcome could leak into the model and make the predictions unrealistically accurate.